



ZIMBABWE

**MINISTRY OF HIGHER AND TERTIARY EDUCATION,
SCIENCE AND TECHNOLOGY DEVELOPMENT**

**HIGHER EDUCATION EXAMINATIONS COUNCIL
(HEXCO)**

NATIONAL CERTIFICATE

IN

**MOTOR CYCLE MECHANICS, PRECISION MACHINING, MOTOR VEHICLE
MECHANICS, DIESEL PLANT FITTING, AUTO-ELECTRICS, MOTOR VEHICLE
BODY REPAIR AND REFINISHING**

SUBJECT: Automotive Engineering Science PAPER NO: 346/13/S09

DURATION: 3 Hours

OCTOBER/NOVEMBER 2019 EXAMINATION

REQUIREMENTS

Pen, pencil, scientific calculator

INSTRUCTIONS TO CANDIDATES

SECTION A: Answer ALL questions (40 marks)

SECTION B: Any five (5) questions (60 marks)

This paper consists of 5 printed pages.

SECTION A

QUESTION 1

- 1.3 Distinguish between a vector quantity and a scalar quantity (2 marks)
- 1.2 Find $\overline{AB}$ where A is the point (2;-1) and B is the point (-1; 3) (2 marks)
- 1.3 A vehicle hauls a trailer at 72km/h when exerting a steady pull of 800N at the tow rope. Calculate the power required (3 marks)
- 1.4 A hydraulic jack has a ram area of 2000mm^2 and a lifting capacity of 30KN. Calculate the compressive stress produced in the ram when the jack is used to 70% of its maximum capacity. (5 marks)
- 1.5 State three (3) motor vehicle components in which friction has a positive effect (3 marks)
- 1.6 Calculate the frictional torque resisting the rotation of a shaft journal 100mm diameter. The load on the journal being 17.5KN and a coefficient of friction between the shaft and the bearing being 0.06. (4 marks)
- 1.7 A propelling force of 600N acts on a body of mass 40kg against a resistance of 100N. Calculate the acceleration of the body. (3 marks)
- 1.8 Calculate work done when a casting is moved 15m along a workshop floor by a force of 7.5KN. (2 marks)
- 1.9 Define
- a) Brake power (1 mark)
 - b) Indicated power (1 mark)
- 1.10 A motor vehicle travels round a bend of 45m radius at 60km/h. What is the angle turned through by the vehicle in one second
- a) in radians (2 marks)
 - b) in degrees (2 marks)
- 1.11 A pushrod has a cross-sectional area of 20mm^2 . At a certain point when operating a valve the force in the rod is 1KN. Determine the compressive stress in the rod. (3 marks)
- 1.12 0.2kg of molten lead at its melting point of 327°C is allowed to solidify at the same temperature and gives out 5200J of heat energy. What is the specific latent heat of fusion of lead. (3 marks)

- 1.13 Copper has a coefficient of linear expansion $17 \times 10^{-6}/K$. Find for copper,
- coefficient of superficial expansion (2 marks)
 - coefficient of volumetric expansion (2 marks)

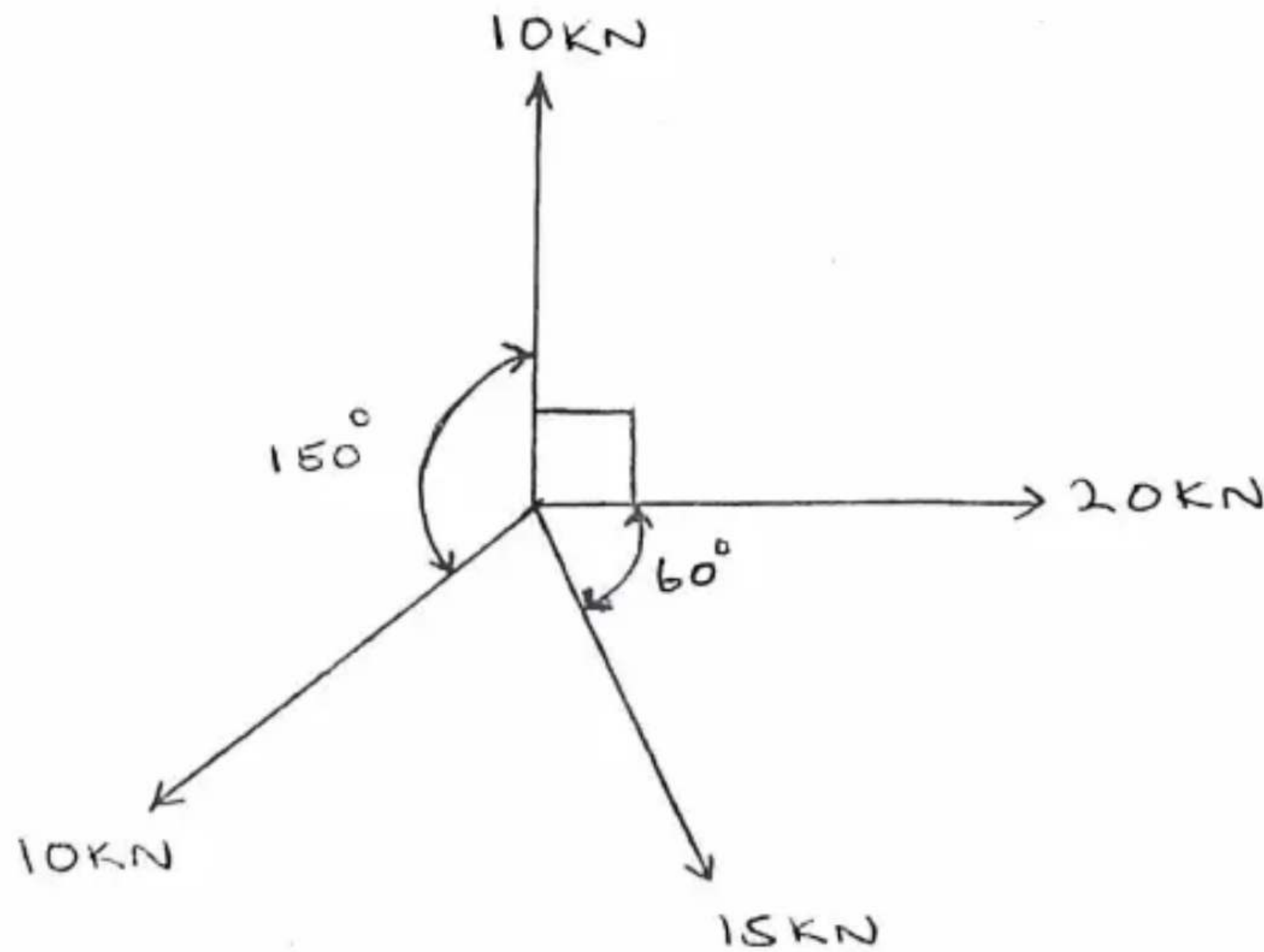
SECTION B

QUESTION 2

- a) A hand operated lifting machine in a repair shop raises an engine of mass 260kg by means of an effort of 210N at the handle. If the effort moves through a distance of 13.5m in raising the engine 450mm.

Calculate:

- the force ratio
 - the movement ratio
 - the efficiency of the machine (6 marks)
- b) Determine by constructing the polygon of forces, the magnitude and direction of the resultant for the system of concurrent coplanar forces shown below (6 marks)



QUESTION 3

A uniform beam 4m long is simply supported at two points A and B. Point A being 0.5m from the left hand end and point B is 1.5m from the right hand end. The beam carries loads of 600N at the left hand end, 800N at its centre and 400N at the right hand end.

- Determine the magnitude of the support reactions at A and B (4 marks)
- At what point should the load of 800N be applied to make the support reactions equal (4 marks)
- A handbrake lever is pivoted at a point 360mm from the handle and the brake pull rod is coupled to the lever 88mm beyond this point. If a force of 90N is applied to the handle of the lever, what will be the pull in the brake rod, assuming that the forces are parallel and perpendicular to the lever. (4 marks)

QUESTION 4

A six cylinder, four stroke engine was coupled to a hydraulic dynamometer for testing. The engine developed maximum power at a speed of 4800 rev/min and the combined static load and spring balance reading was found to be 285N. If the constant K for the dynamometer is 16. Determine.

- the b.p developed by the engine (6 marks)
- the torque transmitted (6 marks)

QUESTION 5

- A vehicle of mass 2 tonnes is being propelled along a level road by a constant tractive effort of 1.25kN. If the rolling resistance amounts to 100N/tonne of the total mass, determine the acceleration of the vehicle. Neglect air resistances (6 marks)
- A motor lorry has a mass of 4Mg and is driven up a hill of gradient 1 in 20 at a constant speed of 36km/h. If the rolling resistance amounts to 75N/tonne. Calculate the tractive effort exerted by the vehicle. (6 marks)

QUESTION 6

Find the quantity of steam at atmospheric pressure that is required to change 0.2kg of ice at -6°C into water at 50°C . The specific latent heat of vaporisation of steam is 2260kJ/kg and the specific latent heat of fusion of ice is 335kJ/kg. The specific heat capacity of ice is 2.1kJ/kg K and the specific heat capacity of water is 4.2kJ/kgK. (12 marks)

QUESTION 7

The following results were obtained during a tensile test to destruction on a specimen of low carbon steel.

Diameter of specimen	=	11.28mm
Gauge length	=	50mm
Load at yield point	=	26KN
Maximum load on specimen	=	47.5KN
Final length at point of fracture	=	67mm
Diameter of specimen at fracture	=	8.2mm

From these results, determine the

- | | |
|-------------------------------------|-----------|
| a) yield stress | (3 marks) |
| b) the tensile strength | (3 marks) |
| c) the percentage elongation | (3 marks) |
| d) the percentage reduction in area | (3 marks) |

QUESTION 8

- a) A motor vehicle gearbox has a first gear ratio of 3.75 to 1. The input shaft of the gearbox runs at 2400rpm with input torque of 150Nm. The efficiency of the gearing is 80%.
Calculate:
- | | |
|----------------------------------|-----------|
| i) the speed of the output shaft | (3 marks) |
| ii) the output torque | (3 marks) |
- b) A rear axle of a motor vehicle is fitted with a crown wheel which has 72 teeth and the corresponding bevel gear has 17 teeth. Determine the rear axle reduction ratio. (6 marks)

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